MIL-S-19500/135A(NAVY)
18 DECEMBER 1968
SUPERSEDING
MIL-S-19500/135(NAVY)
17 October 1960

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, GERMANIUM

TYPE 2N695

1. SCOPE

1.1 Scope. This specification covers the detail requirements for a germanium PNP transistor and is in accordance with MIL-S-19500, except as otherwise specified herein.

1.2 Physical dimensions. See figure 1.

1.3 Maximum ratings.

P_C 1/ T_C = +25° C	P_C 2/ T_A = +25° C	V_{CB}	V_{EB}	V_{CE}	I_C	T_{stg}	T_J
mW	mW	Vdc	Vdc	Vdc	mAdc	°C	°C
150	75	-15	-3.5	-15	50	-65 to +100	+100

1/ Derate linearly 2 mW/° C for T_C >25° C.
2/ Derate linearly 1 mW/° C for T_A >25° C.

1.4 Primary electrical characteristics.

Limits	h_{FE} V_{CE} = -0.3 Vdc I_C = -10 mA	$V_{CE(sat)}$ I_C = -50 mA I_B = 2.5 mA	$t_d + t_r$	t_s	t_f
			nsec	nsec	nsec
Min	25	---	---	---	---
Max	---	1.0	75	100	100

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 General. Requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein defined in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The transistor shall be of the design, construction, physical dimensions shown on figure 1.

3.4 Performance characteristics. Performance characteristics shall be as specified in tables I and

3.5 Marking. The following marking specified in MIL-S-19500 may be omitted from the body of the sistor at the option of the manufacturer:

(a) Country of origin
(b) Manufacturer's designating symbol

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as cified herein.

4.2 Qualification inspection. Qualification inspection shall consist of the examinations and tests speci- in tables I and II.

4.3 Quality conformance inspection. Quality conformance inspection shall consist of groups A and B sections.

4.3.1 Group A inspection. Group A inspection shall consist of the examinations and tests specified in e I.

4.3.2 Group B inspection. Group B inspection shall consist of the examinations and tests specified in e II.

4.4 Methods of examination and test. Methods of examination and test shall be as specified in tables I II.

TABLE I. Group A inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1			10				
isual and mechanical examination	2071			---	---	---	---
Subgroup 2			5				
reakdown voltage, collector to emitter	3011	Bias cond. C I_C = -100 µA		BV_{CES}	-15	---	Vdc
reakdown voltage, collector to base	3001	Bias cond. D I_C = -100 µA		BV_{CBO}	-15	---	Vdc

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TABLE I. Group A inspection (Continued)

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 2 (cont'd.)							
Breakdown voltage, emitter to base	3026	Bias cond. D I_E = -100 µA		BV_{EBO}	-3.5	---	Vdc
Collector to base cutoff current	3036	Bias cond. D V_{CB} = -6.0 Vdc		I_{CBO}	---	-3	µAdc
Open circuit output capacitance	3236	V_{CB} = -6.0 Vdc; I_E = 0; 100 kHz < f < 1 MHz		C_{obo}	---	5	pf
Forward-current transfer ratio	3076	V_{CE} = -0.3 Vdc I_C = -10 mA		h_{FE}	25	---	---
Collector to emitter voltage (saturated)	3071	I_C = -50 mAdc I_B = 2.5 mAdc		V_{CE} (sat)	---	1.0	Vdc
Base emitter voltage	3066	Test cond. B I_B = 0.4 mAdc I_C = 10 mAdc		V_{BE}	0.34	0.44	Vdc
Delay pulse rise time	3251	Test cond. B V_{CC} = -3.5 Vdc R_C = 300 ohms R_B = 5K ohms $V_{BE(1)}$ = -5.4 Vdc $V_{BE(0)}$ = +1.25 Vdc		$t_d + t_r$	---	75	nsec
Storage time	3251	Test cond. B V_{CC} = -3.5 Vdc R_C = 300 ohms R_B = 5K ohms $V_{BE(1)}$ = -5.4 Vdc $V_{BE(2)}$ = +1.25 Vdc		t_s	---	100	nsec
Fall time	3251	Test cond. B V_{CC} = -3.5 Vdc R_C = 300 ohms R_B = 5K ohms $V_{BE(1)}$ = -5.4 Vdc $V_{BE(2)}$ = +1.25 Vdc		t_f	---	100	nsec
Subgroup 3			10				
Collector to base cutoff current	3036	Bias cond. D V_{CB} = -6.0 Vdc T_A = 71° C		I_{CBO}	---	-100	µAdc
Forward-current transfer ratio	3076	V_{CE} = -0.3 Vdc I_C = -10 mAdc T_A = -55° C		h_{FE}	12	---	---

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TABLE II. Group B inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1			10				
Physical dimensions	2066	(See figure 1)		---	---	---	---
Subgroup 2			10				
Solderability	2026			---	---	---	---
Thermal shock (temperature cycling)	1051	Test cond. A		---	---	---	---
Thermal shock (glass strain	1056	Test cond. A		---	---	---	---
Moisture resistance	1021			---	---	---	---
End points:							
Collector to base cutoff current	3036	Bias cond. D V_{CB} = -6.0 Vdc		I_{CBO}	---	4.5	µAdc
Forward-current transfer ratio	3076	I_C = -10 mAdc V_{CE} = -0.3 Vdc		h_{FE}	20	---	---
Subgroup 3			10				
Shock	2016	Nonoperating; 500 G, approx. 1 msec; 5 blows in each orientation: X_1, Y_1, Y_2, and Z_1.		---	---	---	---
Vibration fatigue	2046	Nonoperating		---	---	---	---
Constant acceleration	2006	10,000 G; in each orientation: X_1, Y_1, Y_2, and Z_1.					
End points:							
(Same as subgroup 2)							
Subgroup 4			10				
Terminal strength (lead fatigue)	2036	Test cond. E		---	---	---	---
Subgroup 5			10				
Salt atmosphere (corrosion)	1041			---	---	---	---
End points:							
(Same as subgroup 2)							
Subgroup 6			λ = 5				
High-temperature life (nonoperating)	1031	T_{stg} = 100° C		---	---	---	---

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TABLE II. Group B inspection (Continued)

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 6 (cont'd.) End points: (Same as subgroup 2)							
Subgroup 7 Steady state operation life	1026	P_C = 75 mW V_{CB} = -10 Vdc	λ = 5	---	---	---	---
End points: (Same as subgroup 2)							

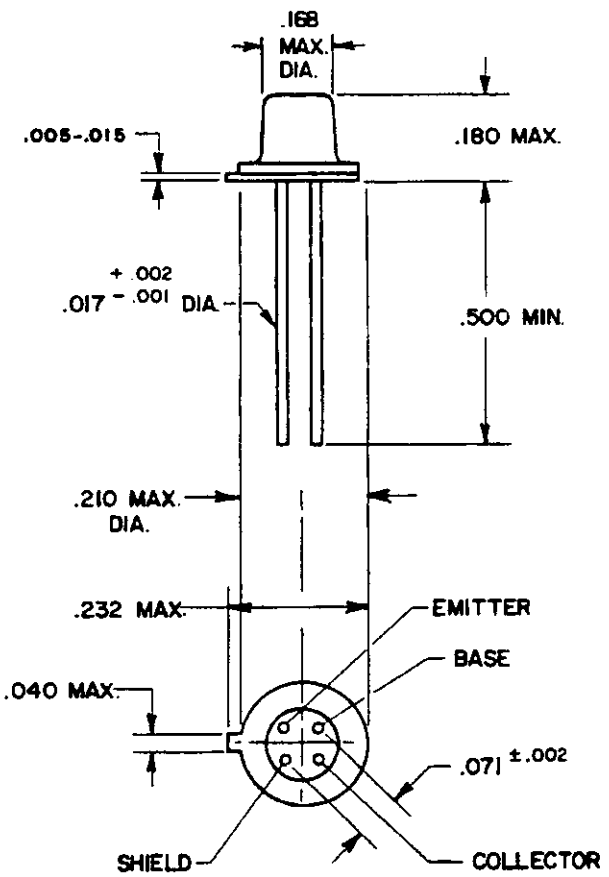


FIGURE 1. Outline dimensions for transistor 2N695.

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5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-S-19500.

6. NOTES

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

:view activity:
Navy - SH
:er activities:
Navy - AS, CG, MC, OS

Preparing activity:
Navy - EC
(Project 5961-N212)

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